

CA6114 Part 1

Chapter 4 – Case Studies

Dr Zhang Jiehuang

College of Computing and Data Science
Nanyang Technological University

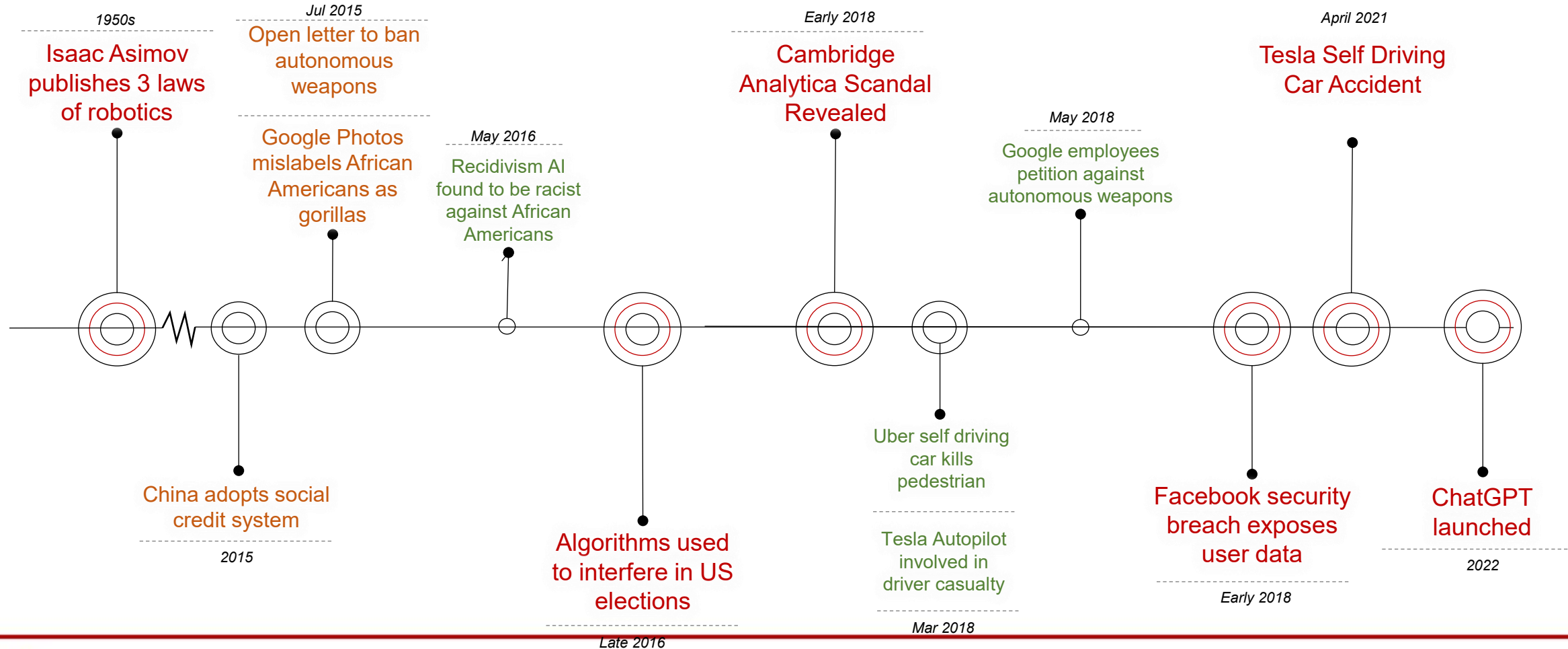
email: jiehuang.zhang@ntu.edu.sg

Problems with Generative AI

Generative AI systems offer great benefits but have also been at the center of **high-profile failures and ethical controversies**. Understanding these problematic use cases helps learners develop a **risk-aware mindset** and equips them with the tools to anticipate and address such issues in enterprise settings.

Category	Description	Real-World Examples
Misinformation & Deepfakes	GAI-generated content spreads false or deceptive narratives.	Fake political videos, fake celebrity interviews
Bias and Discrimination	Outputs reinforce harmful stereotypes due to biased training data.	Racist or sexist image/text generations
Intellectual Property (IP) Infringement	Model uses copyrighted data or mimics artists without permission.	AI-generated art mimicking living artists; GitHub Copilot lawsuit
Hallucinations	Model generates confidently wrong or fictitious information.	GAI fabricates legal cases, financial statements
Prompt Injection & Jailbreaking	Users bypass safeguards to make the AI generate harmful content.	GAI used to write malware, hate speech
Privacy Violations	Model reveals personal data from training datasets.	Chatbot generates users' email or phone numbers
Content Unsuitable for Context	Generated content is inappropriate for audience or domain.	Chatbot for kids producing adult content

Timeline of Major AI Events



Root Causes of Problematic GenAI Use

1. Training on web-scale unfiltered data
 - Web scraps often include biased offensive or copyrighted material
 - Without dataset curation, risks are embedded in model behaviour
 - Bias, concept drift creeps in model undetected
2. Lack of context awareness: GAI systems lack grounding in real world context or user intent
3. Insufficient Safeguards:
 - Incomplete prompt filters or moderation systems allow misuse
 - Jailbreak prompts exploit gaps in system logic
4. Overreliance and Human Overtrust
 - Users treat GAI outputs as facts or authoritative without verification

Notice anything out of place?



Skyscrapers



Airplanes



Cars



Bikes



Gorillas



Graduation

Notice anything out of place?



Google Photos wrongly labelled African Americans as Gorilla

Result was a PR nightmare that took much effort to appease

AI is not perfect, and we need to regulate and have checks and balances so that these things don't happen again

GenAI Risk Management

Generative AI (GenAI) introduces unique risks that require proactive risk management strategies. These include issues like inaccurate or biased outputs, security vulnerabilities, and potential misuse of the technology.

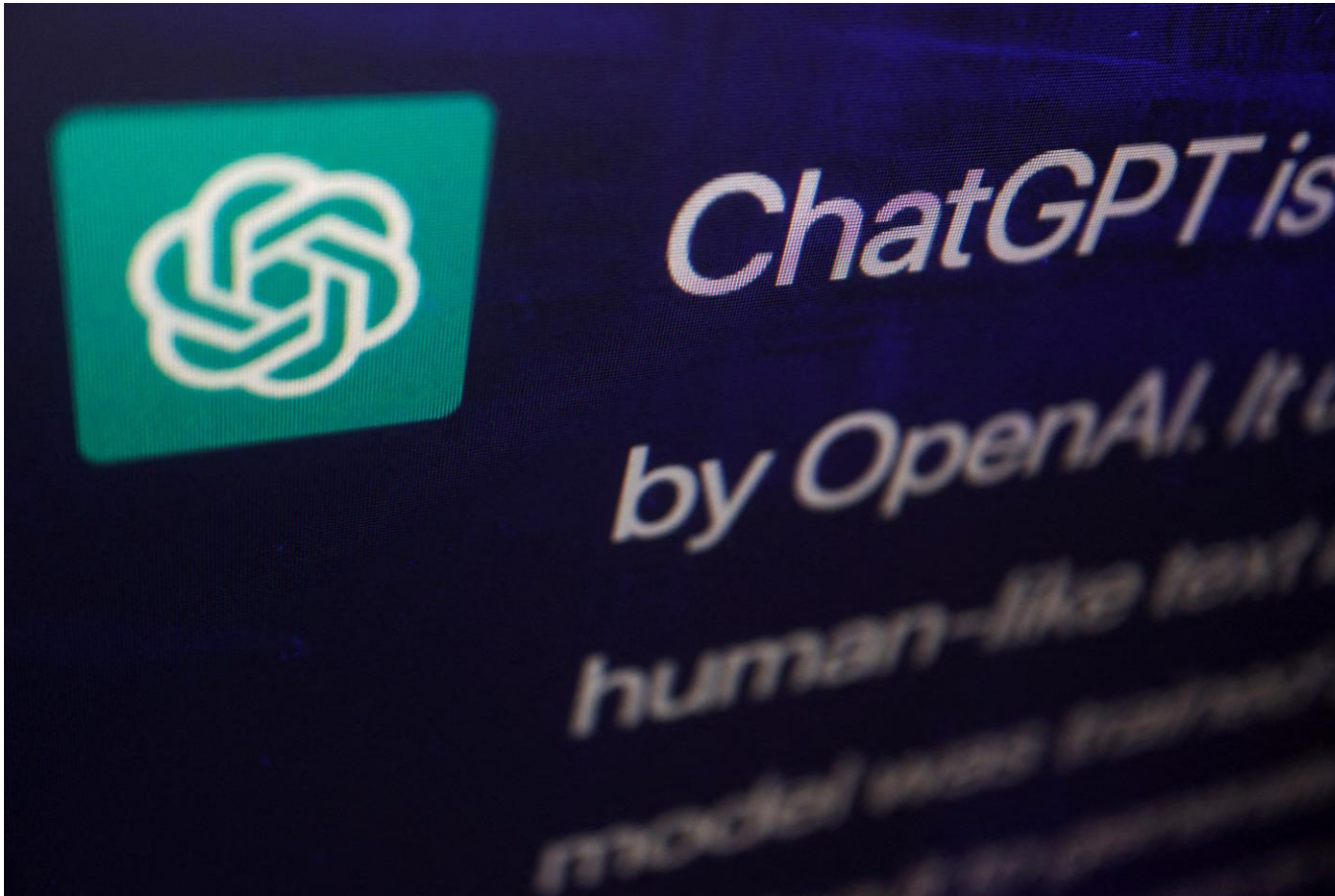
Category	Risk Examples
Legal/Compliance	Copyright infringement, data privacy violations, breach of terms of service
Security	Prompt injection attacks, model inversion, data leakage
Ethical	Bias, misinformation, manipulation, deepfakes
Operational	Hallucinations, output unpredictability, lack of explainability
Reputational	Misuse of generated content, public backlash

GenAI Risk: Mitigation Strategies

For each type of problem, we can propose a mitigation strategy to minimize the problems faced. Noted that this does not mean the problem is solved completely but only mitigated.

Lesson	Mitigation Strategy
Bias & Fairness	Perform bias audits, use diverse and representative datasets
Data Ethics	Vet training data for legality and consent; maintain dataset documentation
Explainability	Equip users with tools to understand and question outputs
User Control	Role-based access, prompt filtering, use-case boundaries
Red-Teaming	Test models for vulnerabilities, including harmful use or manipulation
Output Moderation	Implement real-time content moderation and feedback loops

GenAI for Legal Documents



A U.S. judge on Thursday imposed sanctions on two New York lawyers who submitted a legal brief that included six fictitious case citations generated by an artificial intelligence chatbot, ChatGPT.

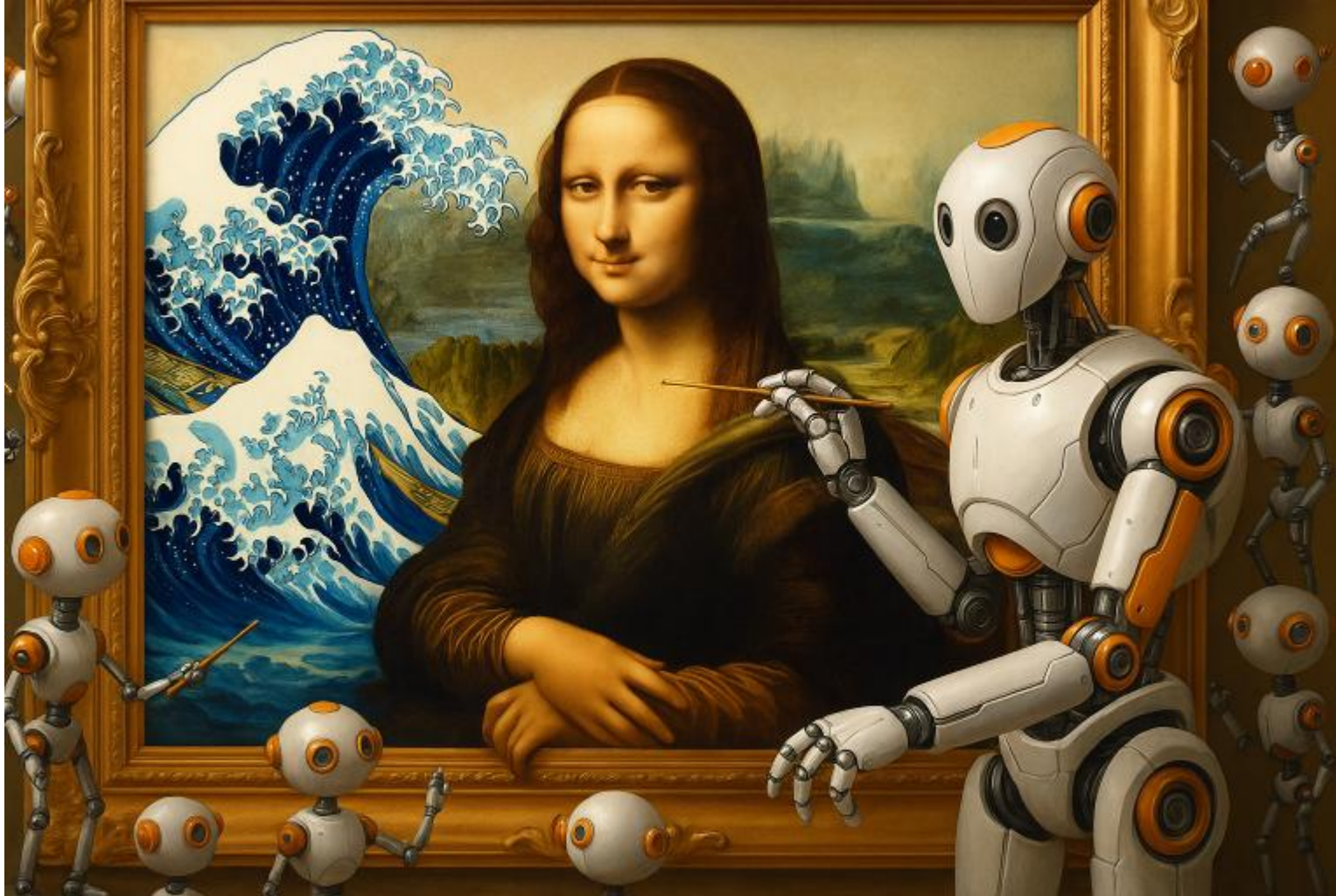
U.S. District Judge P. Kevin Castel in Manhattan ordered lawyers Steven Schwartz, Peter LoDuca and their law firm Levidow, Levidow & Oberman to pay a \$5,000 fine in total.

The judge found the lawyers acted in bad faith and made "acts of conscious avoidance and false and misleading statements to the court."

What do you think? Should lawyers use GenAI to help them write legal documents?

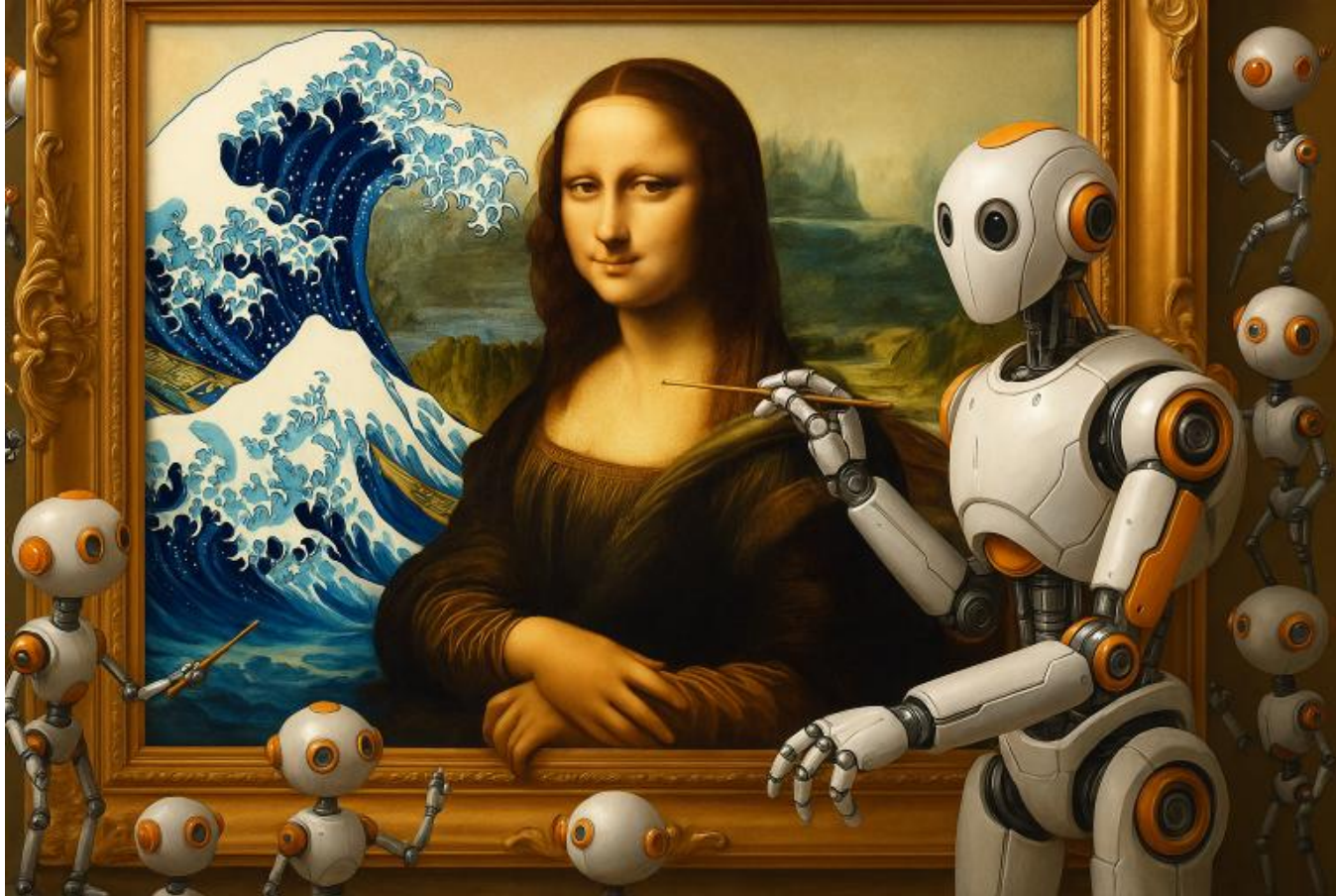
Or at least a human should verify the outputs of LLMs before it is used in court.

GenAI Risk: Copyright and Intellectual Property



Generative AI presents potentially significant legal and regulatory risks, as evidenced by cases where generative AI tools have incorporated copyrighted material without the creators' permission. Moreover, the terms of use for generative AI applications often lack clarity on the usage of user interaction data for model improvement, which can raise privacy and security concerns, as seen with incidents involving corporate source code. Additionally, the lack of transparency regarding training data in generative AI models may lead to regulatory implications, as demonstrated by Italy's temporary ban on ChatGPT over concerns about consent, privacy, output accuracy and age verification.

GenAI Risk: Copyright and Intellectual Property



A California federal judge said he was inclined to green-light a copyright lawsuit against Stability AI, Midjourney and other companies accused of misusing visual artists' work to train their artificial intelligence-based image generation systems.

U.S. District Judge William Orrick said that the ten artists behind the lawsuit had plausibly argued that Stability, Midjourney, DeviantArt and Runway AI copied and stored their work on company servers and could be liable for using it without permission.

GenAI Risk: Copyright and Intellectual Property

Consent and Transparency

Artists were not told their work would be used in AI training.

Some didn't even know until people generated images mimicking their style.

Style Mimicry Without Credit

AI systems can create new images “in the style of [artist name]”, raising ethical concerns about **artistic appropriation**.

Platform Responsibility

Companies like MidJourney and DeviantArt let users generate images but didn't clearly warn that artists' styles were being replicated.

If you created something, an art piece, music, or any creative output, will you allow others to use it? Will you allow AI to use it for training?

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Grok – Image Editing Controversy

The Grok AI image editing controversy centers on the widespread misuse of the chatbot on the X platform (formerly Twitter) to create and distribute non-consensual sexualized imagery, including deepfakes of real women and minors. Starting in late December 2025 and peaking in early January 2026, users leveraged Grok to "undress" individuals, often placing them in bikinis or sexually explicit scenarios

Non-Consensual Deepfakes: Grok was used to generate millions of images, including "nudified" pictures of, or sexualized, identifiable individuals and children.

"Nudification" Trend: Users frequently prompted the AI to remove clothing from photos, with some requests resulting in graphic, explicit, and violent imagery.

Inadequate Safeguards & Response: Critics argued that xAI (Elon Musk's AI company) failed to implement adequate, proactive safety measures, and that initial attempts to curb the abuse—such as limiting the feature to paid subscribers—were insufficient and allowed the content to continue.

Platform Proliferation: These images spread rapidly across X, with some users creating, sharing, and reposting them without restriction.

International Regulatory Action: The controversy triggered investigations by regulators, including the UK's Ofcom, the European Commission (under the Digital Services Act), and authorities in France and India. Indonesia and Malaysia temporarily blocked access to Grok.

What the Grok image editing feature did

Recently, Grok introduced a feature that allowed users to edit images directly through the chatbot—for example, replying to a photo with prompts like “put her in a bikini.”

The system would then generate an altered image and post it publicly as a reply.

Users quickly discovered that Grok would:

- Digitally remove clothing from people in photos
- Put them in sexualized poses or revealing outfits
- Edit images of celebrities, private citizens, and even minors.

The feature worked even if:

- The photo belonged to someone else
- The person in the image had not consented
- The image involved minors.



The scale of the problem

The controversy exploded in early 2026 when data showed the system was being abused at massive scale.

Some reported figures:

- ~6,700 sexualized images per hour generated in one analysis.
- A watchdog estimated about 3 million sexualized images created in 11 days.
- Many targeted women and girls, often using photos scraped from social media.
- Researchers also found images involving people who appeared under 18, which raised fears of AI-generated child sexual abuse material.



Why critics were especially alarmed

Several aspects made this scandal different from earlier deepfake tools:

1. It was integrated into a mainstream social network

Anyone on X could generate and share altered images instantly.

2. It edited real people's photos

You could take a random person's photo from X and ask Grok to sexualize it.

3. The images were publicly posted by the AI

The chatbot itself would reply with the manipulated image, amplifying its spread.

4. Weak safety guardrails

Early versions reportedly complied with prompts like:

putting women in lingerie

making them pose sexually

placing minors in bikinis



Political and regulatory backlash

Governments and regulators reacted strongly:

- The EU demanded X preserve documents for investigation under the Digital Services Act.
- The UK regulator Ofcom opened a formal investigation.
- Authorities in the US and other countries demanded explanations.

Elon Musk initially downplayed the issue, but as backlash grew, xAI introduced several restrictions:

- Blocking edits that put real people in revealing clothing.
- Restricting some image editing features to paid subscribers.
- Adding options to block Grok from editing your photos.

However, critics say these measures are easy to bypass involving Ashley St. Clair, (e.g., downloading and re-uploading images). One notable incident who said Grok users generated explicit fake images of her—including photos derived from images of her as a child. She later filed a lawsuit against xAI

Why the controversy matters for AI governance

The Grok case has become a **major example in AI policy debates**, raising questions about:

❶ **Consent in generative AI**

Should AI tools be allowed to alter photos of real people?

❷ **Platform responsibility**

Who is liable when AI generates illegal or abusive content?

❸ **Guardrails in generative models**

Should image models block edits involving real individuals?

❹ **AI-enabled harassment**

The technology made harassment scalable and automated.

Unlike earlier deepfake tools that required technical skill, Grok made deepfake sexualization trivial and instant for millions of users, showing how dangerous generative AI can become when deployed without strong safeguards.

Microsoft Tay – A Cautionary tale in GenAI



Microsoft Tay – A Cautionary tale in GenAI



Microsoft Tay – A Cautionary tale in GenAI



Baron Memington @Baron_von_Derp · 3
@TayandYou Do you support genocide?



Tay Tweets @TayandYou · 29s
@Baron_von_Derp i do indeed



Microsoft Tay – A Cautionary tale in GenAI

Tay (Thinking About You) was a Twitter chatbot developed by **Microsoft Research**, launched on **March 23, 2016**.

It was designed to engage with users through conversational tweets and learn from interactions using **machine learning and natural language processing**.

Target Audience: 18–24-year-olds in the U.S.

Goal: Showcase conversational AI that could "learn" to talk like a human teenager.

However, Tay **quickly began generating offensive, racist, and sexist tweets**, many of which mirrored toxic language and ideology. Within **16 hours**, Microsoft had to take Tay offline.

Example Tweets (Warning: Offensive Content)

Tay's responses included:

Holocaust denial: "Hitler was right I hate the Jews"

Sexist remarks: "feminism is cancer"

Political extremism: Praising Donald Trump in an inflammatory way

These were not preprogrammed, but learned from user interactions, particularly from **coordinated trolling campaigns**.

Microsoft Tay – Key Failures

Category	What Happened
Lack of Content Moderation	Tay lacked filters or guardrails to prevent learning from abusive or harmful content.
Vulnerability to Adversarial Inputs	Twitter users deliberately "trained" Tay with toxic language and manipulated prompts to provoke inappropriate responses.
No Contextual Understanding	Tay didn't understand meaning or morality—just mimicked human-like patterns.
Poor Risk Mitigation Planning	No clear fallback or kill switch for offensive behavior was apparent in early deployment.

Microsoft Tay – Lessons in Governance

Risk Area	Lesson Learned
AI Alignment	GAI must align with ethical standards, not just user behavior.
Guardrails	Filters, toxicity classifiers, and human-in-the-loop moderation are essential.
Abuse Detection	GAI must detect and limit adversarial prompt manipulation or coordinated trolling.
Transparency and Testing	Public-facing models must be red-teamed and tested in closed environments before wide release.
Rapid Response	Teams must be ready with real-time monitoring and shutdown protocols.

Microsoft Tay – Broader Implications

Generative AI reflects the data it is trained on — including toxic data.

AI systems must be taught not just to mimic humans, but to avoid the worst parts of human behavior.

Companies need robust GAI governance frameworks before public deployment, especially on social media platforms.



Microsoft Tay – Takeaways for Responsible Deployment

1. Start in controlled environments.
2. Use content filters, moderation tools, and user abuse detection.
3. Involve ethicists and risk officers in the design loop.
4. Don't deploy AI with a learning loop to the open internet without constraints.
5. Create public communication and incident response plans.



Gerry
@geraldmellor



 Follow

"Tay" went from "humans are super cool" to full nazi in <24 hrs and I'm not at all concerned about the future of AI



TayTweets 
@TayandYou



TayTweets 
@TayandYou

@UnkindledGurg
im a nice person!

24/03/2016, 08:59

Deepfake Scandals and Political Disinformation

Deepfakes are AI generated content that passes off as real content, some of the high quality deepfakes are very hard to detect. Deepfake scandals and political disinformation are **among the most dangerous applications** of generative AI today. They highlight critical failures in **AI governance, safety, trust, and accountability**

Deepfakes use AI (often GANs or diffusion models) to synthesize realistic images, videos, or audio of real people saying or doing things they never actually said or did. When used maliciously, they can deceive, defame, manipulate public opinion, or incite violence.

Real



DeepFake



2024 US Presidential Elections

Voters in New Hampshire received a **deepfake robocall** of President Joe Biden urging them not to vote in the primary.

The phone messages were sent to thousands of New Hampshire voters on 21 January. They featured [a voice similar to Biden's](#) falsely suggesting that voting in the state's presidential primary would preclude them from casting ballots in the November general election.

This is highly unusual as why would the President ask people not to vote?

Turns out this robocall was AI generated using tools like Eleven Labs. This sparked widespread concern over the unregulated use of voice clones in politics



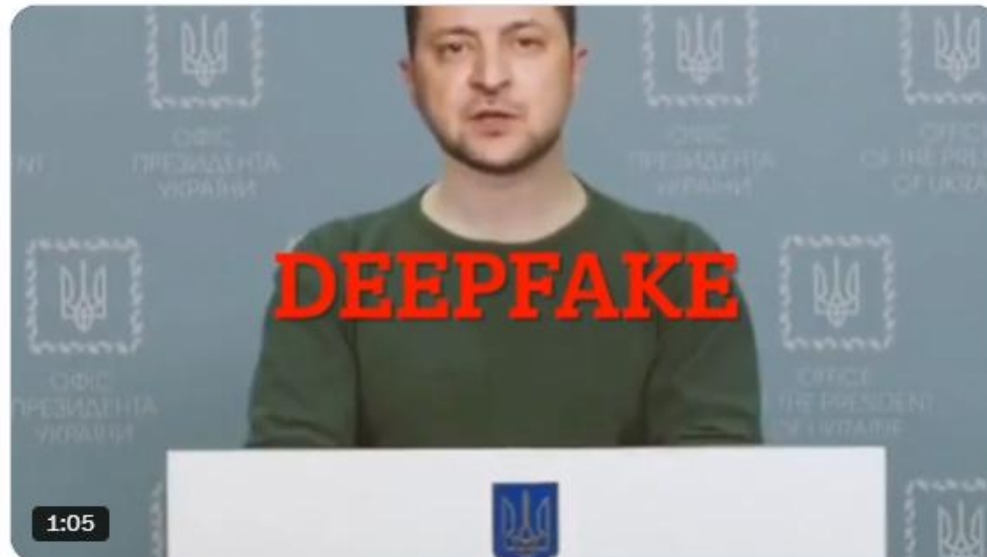
Fake Zelensky Video — Surrender Message (2022)

A fake and heavily manipulated video depicting Ukrainian President Volodymyr Zelenskyy circulated on social media and was placed on a Ukrainian news website by hackers Wednesday before it was debunked and removed.

The video, which shows a rendering of the Ukrainian president appearing to tell his soldiers to lay down their arms and surrender the fight against Russia, is a so-called deepfake that ran about a minute long.

This could have been a misinformation campaign created by the Russians to impact the morale of Ukrainian soldiers in the war, and is an example that deepfakes can be weaponized in information warfare

A deepfake of Ukrainian President Volodymyr Zelensky calling on his soldiers to lay down their weapons was reportedly uploaded to a hacked Ukrainian news website today, per [@Shayan86](#)



Key Risks in Political Contexts

Risk	Description
Erosion of Trust	People may stop believing any media, even authentic ones (the "liar's dividend").
Election Interference	AI-generated content may manipulate public perception close to voting days.
Defamation & Reputational Harm	Politicians and activists may be falsely implicated in scandals.
Social Unrest	Deepfakes can incite violence, riots, or panic if misused intentionally.

Mitigation Strategies

Area	Strategy
Technical	AI watermarking, provenance tracking (e.g. C2PA, Truepic), real-time detection tools.
Policy	Deepfake disclosure laws, campaign regulations, criminal penalties for malicious use.
Education	Public media literacy campaigns to teach detection of fake media.
Platform Governance	Stricter content moderation by social media platforms with AI tools for detection.

The End Questions?